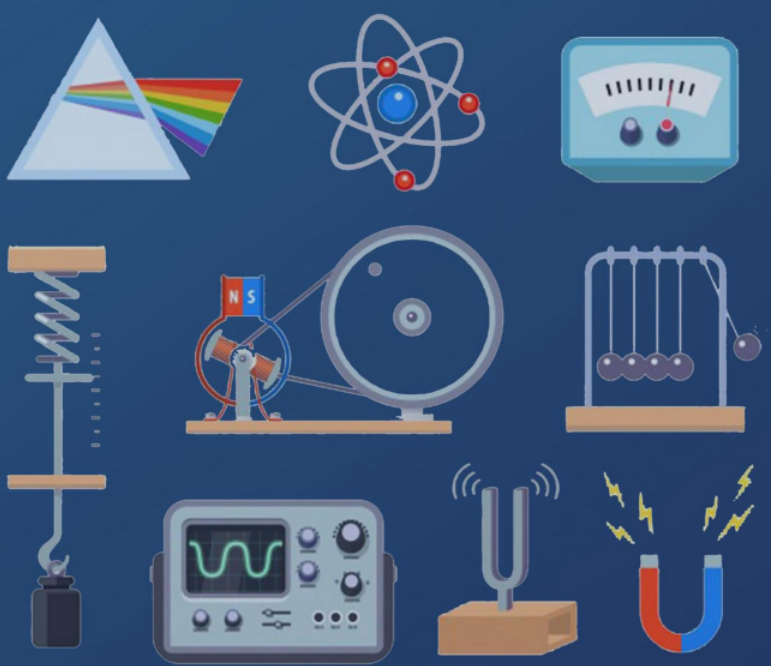
PHYSICS

Simple Harmonic Motion



SHEET

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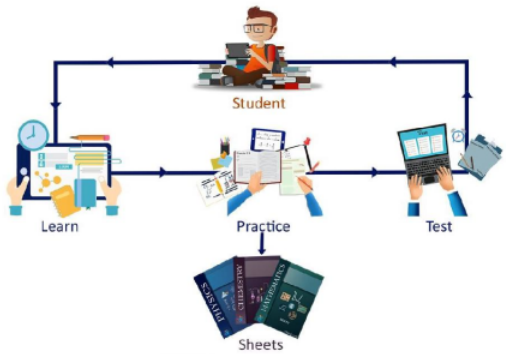
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 3
5.	Exercise 2	Page - 5
6.	Exercise 2A	Page - 8
7.	Answer Key	Page - 13

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction to SHM	1-4	1,2		
2	Phasor Diagram	5	8,9	1	
3	Velocity and Acceleration Analysis	6	3,10,11	2-4	1-4
4	Understanding Graphs	7-10		5,6	16
5	Energy in SHM	11,12	4,5,12	7,8	5,6,17
6	Spring Block System	13-15	13,14	9,10	7,8,18
7	Variation in Parameters in SHM		7		9
8	Two Blocks SHM		6		
9	Simple Pendulum	16	15	11	
10	Physical Pendulum				
11	Miscelleneous Angular SHM			12	
12	Composition of SHM	17		13	10,11,14, 15
13	Damped SHM	18			
14	Forced Oscillations and Resonance				
15	After complete chapter			14,15	12,13,19

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to SHM

Q1. The equation of motion of a particle executing S.H.M. where letters have usual meaning is :

(A) $\frac{d^2x}{dt^2} = -\frac{k}{m}x$

(B) $\frac{d^2x}{dt^2} = +\omega^2 x$

(C) $\frac{d^2x}{dt^2} = -\omega^2 x^2$

(D) $\frac{d^2x}{dt^2} = -kmx$

Q2. The equation of a simple harmonic motion is $x = 0.34\cos(3000t + 0.74)$. Where x and t are in mm and sec. respectively.

The frequency of the motion is :

(A) 3000 (B) $3000/2\pi$

(C) $0.74/2\pi$ (D) $3000/\pi$

Q3. The equation of motion of a particle executing SHM is $\left(\frac{d^2x}{dt^2}\right) + kx = 0$. The time period of the particle will be :

(A) $2\pi/\sqrt{k}$ (B) $2\pi/k$

(C) $2\pi k$ (D) $2\pi\sqrt{k}$

Q4. The angular frequency of motion whose equation is $4\frac{d^2y}{dt^2} + 9y = 0$ is

(y = displacement and t = time)

(A) $\frac{9}{4}$ (B) $\frac{4}{9}$

(C) $\frac{3}{2}$ (D) $\frac{2}{3}$

Phasor Diagram

Q5. The phase of a particle in SHM on X axis at time t is $\pi/6$. The following inference is drawn from this:

(A) The particle is at $x = a/2$ and moving in + X-direction

(B) The particle is at $x = a/2$ and moving in – X-direction

(C) The particle is at $x = -a/2$ and moving in + X-direction

(D) The particle is at $x = -a/2$ and moving in – X-direction

Velocity and Acceleration Analysis

Q6. The displacement of a particle in S.H.M. is indicated by equation $y = 10\sin(20t + \pi/3)$ where y is in metres. The value of time period of vibration will be (in seconds) :

(A) $10/\pi$ (B) $\pi/10$

(C) $2\pi/10$ (D) $10/2\pi$

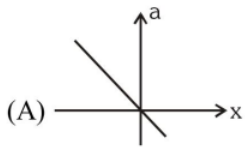
Understanding Graphs

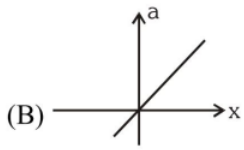
Q7. If the maximum velocity of a particle in SHM is v_0. then its velocity at half the amplitude from position of rest will be :

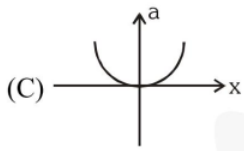
(A) $v_0/2$ (B) v_0

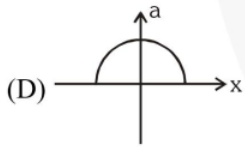
(C) $v_0\sqrt{3/2}$ (D) $v_0\sqrt{3}/2$

Q8. The variation of acceleration (a) and displacement (x) of the particle executing SHM is indicated by the following curve :

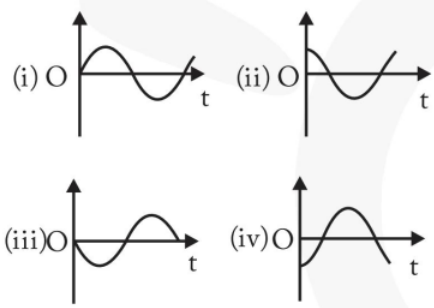
(A) 

(B) 

(C) 

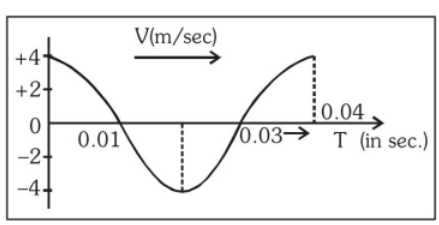
(D) 

Q9. The displacement of a particle in S.H.M. is $x = a \sin\omega t$. Which of the following graph between displacement and time is correct :



(A) (i) (B) (ii) (C) (iii) (D) (iv)

Q10. The velocity-time diagram of a harmonic oscillator is shown in the adjoining figure. The frequency of oscillation is :



(A) 25 Hz (B) 50 Hz
(C) 12.25 Hz (D) 33.3 Hz

Energy in SHM

Q11. The total energy of a harmonic oscillator of mass 2kg is 9 joules. If its potential energy at mean position is 5 joules, its K.E. at the mean position will be :

(A) 9 J (B) 14 J
(C) 4 J (D) 11 J

Q12. The value of total mechanical energy of a particle in S.H.M. is :

(A) Always constant

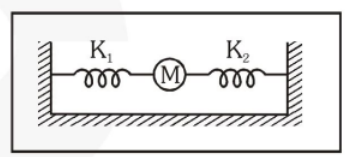
(B) Depend on time

(C) $\frac{1}{2}\ kA^2\cos^2(\omega t+\phi)$

(D) $\frac{1}{2}mA^2\cos^2(\omega t+\phi)$

Spring Block System

Q13. In the adjoining figure the frequency of oscillation for a mass m will be proportional to :



(A) K_1K_2 (B) $K_1 + K_2$
(C) $\sqrt{K_1+K_2}$ (D) $\sqrt{1/K_1+K_2}$

Q14. A horizontal spring is connected to a mass M. It executes simple harmonic motion. When the mass M passes through its mean position, an object of mass m is put on it and the two move together. The ratio of frequencies before and after will be-

(A) $\left(1+\frac{m}{M}\right)^{1/2}$ (B) $\left(1+\frac{m}{M}\right)$

(C) $\left(\frac{M}{M+m}\right)^{1/2}$ (D) $\left(\frac{M}{M+m}\right)$

Q15. A spring of force constant k is cut into two pieces such that one piece is double the length of the other. Then the long piece will have a force constant of :-

(A) $\frac{2}{3}k$ (B) $\frac{3}{2}k$

(C) 3k (D) 6k

Simple Pendulum

Q16. A bob is attached to a long, light string of length 1m. The string is deflected by 3° initially with respect to vertical. The value of θ at any time t after the bob released can be approximately written as (Use $g = \pi^2$)

(A) $3° \cos \pi t$

(B) $3° \sin \pi t$

(C) $3° \sin (\pi t + \frac{\pi}{6})$

(D) $3° \cos (\pi t + \frac{\pi}{6})$

Composition of SHM

Q17. The displacement of a particle varies with time according to the relation $y = a \sin \omega t + b \cos \omega t$.

(A) The motion is oscillatory but not S.H.M.

(B) The motion is S.H.M. with amplitude $a + b$.

(C) The motion is S.H.M. with amplitude $a^2 + b^2$.

(D) The motion is S.H.M. with amplitude $\sqrt{a^2 + b^2}$

Damped SHM

Q18. The amplitude of a damped oscillator becomes half in one minute. The amplitude after 3 minutes will be $\frac{1}{x}$ times of the original. Determine value of x.

(A) 4 (B) 8 (C) 16 (D) 3

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A necessary and sufficient condition for SHM is

(A) constant period.

(B) constant acceleration.

(C) proportionality between acceleration and displacement from equilibrium position.

(D) proportionality between the restoring force and displacement from equilibrium position.

Q2. The rotation of earth about its axis is :-

(A) periodic motion.

(B) simple harmonic motion.

(C) Oscillatory motion.

(D) Vibration

Q3. A particle is executing SHM between points $-X_m$ and X_m, as shown in figure-I. The velocity V(t) of the particle is partially graphed and shown in figure-II. Two points A and B corresponding to time t_1 and time t_2 respectively are marked on the V(t) curve.

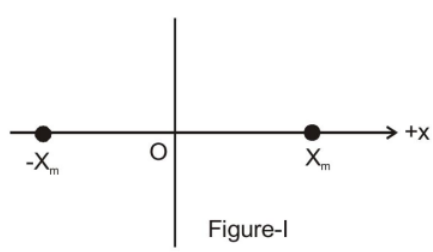


Figure-I

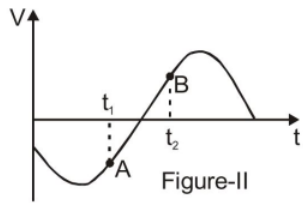


Figure-II

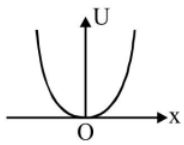
(A) At time t_1 , it is going towards X_m.

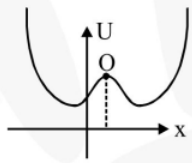
(B) At time t_1, its speed is decreasing.

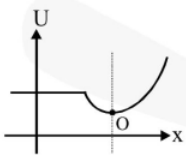
(C) At time t_2, its position lies in between $-X_m$ and O.

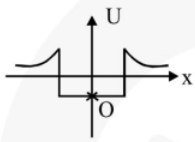
(D) The phase difference $\Delta\phi$ between points A and B must be expressed as $90° < \Delta\phi < 180°$.

Q4. Potential energy vs displacement graph is given for different situations, then for a small displacement about position 'O' in which of the following motion is S.H.M. :-

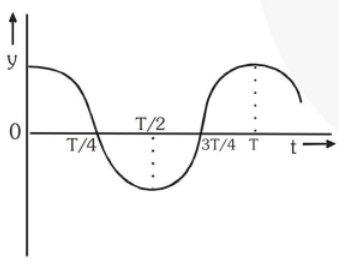
(A) 

(B) 

(C) 

(D) 

Q5. The displacement-time graph of a particle executing SHM is shown.



Which of the following statements is/are true?

(A) The velocity is maximum at t = T/2

(B) The acceleration is maximum at t = T

(C) The force is zero at t = 3T/4

(D) The potential energy equals the oscillation energy at t = T/2.

Q6. Two particles are in SHM with same amplitude A and same angular frequency ω.

At time t = 0, one is at $x = +\frac{A}{2}$ and other is at $x = -\frac{A}{2}$. Both are moving in same direction.

(A) Phase difference between the two particle is $\frac{\pi}{3}$

(B) Phase difference between the two particle is $\frac{2\pi}{3}$

(C) They will collide after time $t = \frac{\pi}{2\omega}$

(D) They will collide after time $t = \frac{3\pi}{\omega}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q7. A body of mass m attached to a spring which is oscillating with time period 4 seconds. If the mass of the body is increased by 4 kg, its time period increases by 2 sec. Determine value of initial mass m in Kg.

SUBJECTIVE TYPE

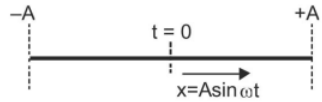
Q8. If two S.H.M.'s are represented by equations

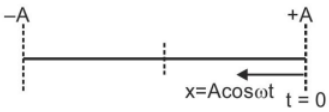
$$y_1 = 10\sin\left[3\pi t + \frac{\pi}{4}\right]$$

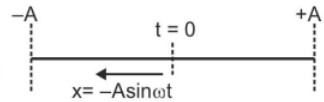
and $y_2 = 5\left[\sin(3\pi t) + \sqrt{3}\cos(3\pi t)\right]$

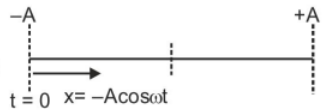
then find the ratio of their amplitudes and phase difference in between them.

Q9. What will be the equation of displacement in the following different conditions ?

(i) 

(ii) 

(iii) 

(iv) 

Q10. An object performs S.H.M. of amplitude 5 cm and time period 4 s. If timing is started when the object is at the centre of the oscillation i.e., x = 0 then calculate.

(a) Frequency of oscillation

(b) The displacement at 0.5 sec.

(c) The maximum acceleration of the object.

(d) The velocity at a displacement of 3 cm.

Q11. Amplitude of a harmonic oscillator is A, when velocity of particle is half of maximum velocity, then determine position of particle.

Q12. In case of simple harmonic motion –

(a) What fraction of total energy is kinetic and what fraction is potential when displacement is one half of the ampli-tude.

(b) At what displacement the kinetic and potential energies are equal.

Q13. A spring of force constant k is cut into two unequal parts ℓ_1 and ℓ_2, where $\frac{\ell_1}{\ell_2} = \frac{n_1}{n_2}$ then determine force constant of each part.

Q14. A block of mass m is suspended from a spring of spring constant k and released from rest at natural length. Find the amplitude of S.H.M.

Q15. The length of a second's pendulum at the surface of earth is 1m. Determine the length of second's pendulum at the surface of moon.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. A particle executes simple harmonic motion between x = –A and x = +A. The time taken for it to go from O to $\frac{A}{2}$ is T_1 and to go from $\frac{A}{2}$ to A is T_2, then :-

(A) $T_1 < T_2$ (B) $T_1 > T_2$

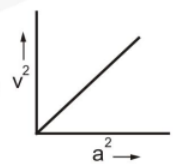
(C) $T_1 = T_2$ (D) $T_1 = 2T_2$

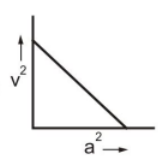
Q2. A particle performing S.H.M. is found at its equilibrium at t = 1 s and it is found to have a speed of 0.25 m/s at t = 2 s. If the period of oscillation is 6s Calculate amplitude of oscillation

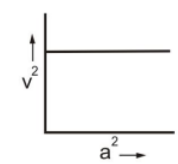
(A) $\frac{3}{2\pi}$ m (B) $\frac{3}{4\pi}$ m

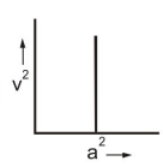
(C) $\frac{6}{\pi}$ m (D) $\frac{3}{8\pi}$ m

Q3. A mass M is performing linear simple harmonic motion, then correct graph for acceleration 'a' and corresponding linear velocity 'v' is

(A) 

(B) 

(C) 

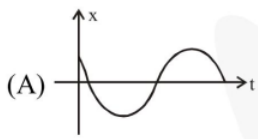
(D) 

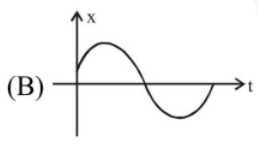
Q4. A particle performs SHM with a period T and amplitude a. The mean velocity of the particle over the time interval during which it travels a distance a/2 from the extreme position is

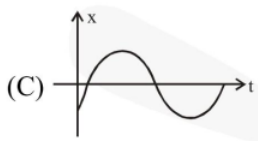
(A) a/T (B) 2a/T

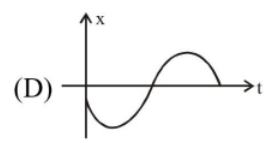
(C) 3a/T (D) a/2T

Q5. A particle performing S.H.M. about mean position x = 0 and at t = 0 it is at position $x = \frac{A}{\sqrt{2}}$ and moving towards the origin. Then which of the following is its possible graph between position (x) and time (t)

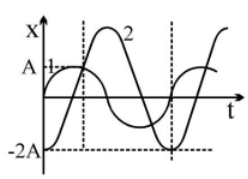
(A) 

(B) 

(C) 

(D) 

Q6. The oscillations represented by curve 1 in the graph are expressed by equation x = A sinωt. The equation for the oscillations represented by curve 2 is expressed as:



(A) x = 2A sin (ωt – π/2)

(B) x = 2A sin (ωt + π/2)

(C) x = – 2A sin (ωt – π/2)

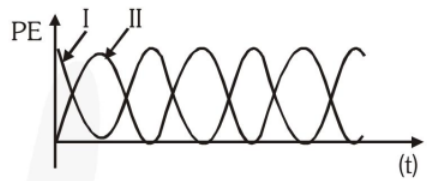
(D) x = A sin (ωt – π/2)

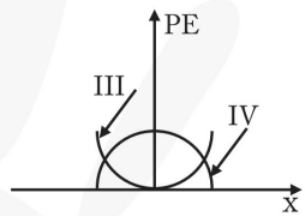
Q7. The total vibrational energy of a particle in S.H.M. is E. Its kinetic energy at half the amplitude from mean position will be :

(A) E/2 (B) E/3

(C) E/4 (D) 3E/4

Q8. For a particle executing SHM the displacement x given by x = Acosωt. Identify the graph which represents the variation of potential energy (PE) as a function of time t and displacement x :-

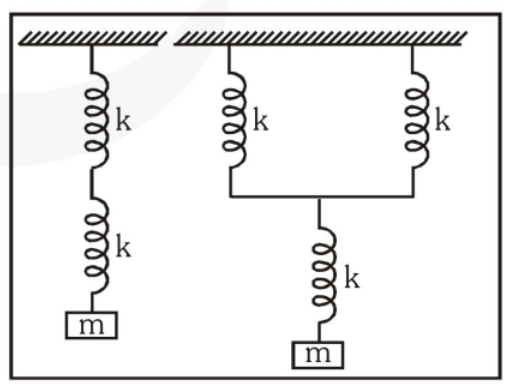




(A) I, III (B) II, IV

(C) II, III (D) I, IV

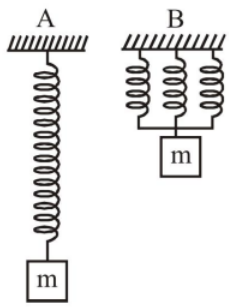
Q9. Some springs are combined in series and parallel arrangement as shown in the figure and a mass m is suspended from them. The ratio of their frequencies will be :



(A) 1 : 1 (B) 2 : 1

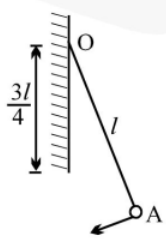
(C) $\sqrt{3}$: 2 (D) 4 : 1

Q10. The springs in figure A and B are identical but length of spring in A is three times than the length of each spring in B. The ratio of period T_A/T_B is :-



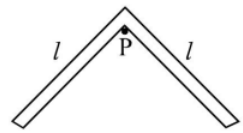
(A) $\sqrt{3}$ (B) 1/3
(C) 3 (D) $1/\sqrt{3}$

Q11. A small bob attached to a light inextensible thread of length *l* has a periodic time T when allowed to vibrate as a simple pendulum. The thread is now suspended from a fixed end O of a vertical rigid rod of length $\frac{3l}{4}$ (as in figure). If now the pendulum performs periodic oscillations in this arrangement, the periodic time will be



(A) $\frac{3T}{4}$ (B) $\frac{T}{2}$
(C) T (D) 2T

Q12. A system of two identical rods (L-shaped) of mass *m* and length *l* are resting on a peg P as shown in the figure. If the system is displaced in its plane by a small angle θ, find the period of oscillations:



(A) $2\pi\sqrt{\frac{2l}{3g}}$ (B) $2\pi\sqrt{\frac{2\sqrt{2}l}{3g}}$
(C) $2\pi\sqrt{\frac{2l}{3g}}$ (D) $3\pi\sqrt{\frac{l}{3g}}$

Q13. Two simple harmonic motions $y_1 = A\sin\omega t$ and $y_2 = A\cos\omega t$ are superimposed on a particle of mass m. The total mechanical energy of the particle is:

(A) $\frac{1}{2}m\omega^2A^2$ (B) $m\omega^2A^2$
(C) $\frac{1}{4}m\omega^2A^2$ (D) zero

Q14. The time taken by a particle performing SHM on a straight line to pass from point A to B where its velocities are same is 2 seconds. After another 2 seconds it returns to B. The time period of oscillation is (in seconds)

(A) 2 (B) 8
(C) 6 (D) 4

Q15. The period of oscillation of simple pendulum of length L suspended from the roof of the vehicle which moves without friction, down an inclined plane of inclination α, is given by :-

(A) $2\pi\sqrt{\frac{L}{g\cos\alpha}}$ (B) $2\pi\sqrt{\frac{L}{g\sin\alpha}}$
(C) $2\pi\sqrt{\frac{L}{g}}$ (D) $2\pi\sqrt{\frac{L}{g\tan\alpha}}$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A horizontal plank has a rectangular block placed on it. The plank starts oscillating vertically and simple harmonically with an amplitude of 40 cm. The block just loses contact with the plank when the latter is at momentary rest. Then :

(A) the period of oscillation is $\left(\frac{2\pi}{5}\right)$

(B) the block weighs double of its weight, when the plank is at one of the positions of momentary rest

(C) the block weighs 0.5 times its weight on the plank halfway up

(D) the block weighs 1.5 times its weight on the plank halfway down

Q2. For a SHM with given angular frequency, two arbitrary initial conditions are necessary and sufficient to determine the motion completely. These initial conditions may be-

(A) Amplitude and initial phase

(B) Amplitude and total energy of oscillation

(C) Initial phase and total energy of oscillation

(D) Initial position and initial velocity

Q3. For a body executing SHM with amplitudes A, time period T, max velocity v_{max} and phase constant zero, which of the following statements are correct?

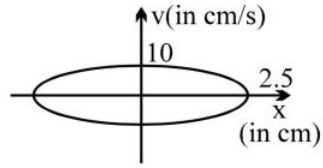
(A) At $y = (A/2)$, $v > (v_{max}/2)$

(B) $v = (v_{max}/2)$ for $y > (A/2)$

(C) For $t = (T/8)$, $y > (A/2)$

(D) For $y = (A/2)$, $t < (T/8)$

Q4. The figure shows a graph between velocity and displacement (from mean position) of a particle performing SHM:



(A) The time period of the particle is 1.57s

(B) The maximum acceleration will be 40 cm/s^2

(C) The velocity of particle is $2\sqrt{21}$ cm/s when it is at a distance 1 cm from the mean position.

(D) None of these

Q5. The amplitude of a particle executing SHM about O is 10 cm . Then :

(A) when the kinetic energy is 0.64 times of its max. kinetic energy its displacement is 6 cm from O

(B) when the displacement is 5 cm from O its kinetic energy is 0.75 times its maximum kinetic energy

(C) Its total energy of SHM at any point is equal to its maximum kinetic energy

(D) Its speed is half the maximum speed when its displacement is half the maximum displacement

Q6. A system is oscillating with undamped simple harmonic motion. Then the

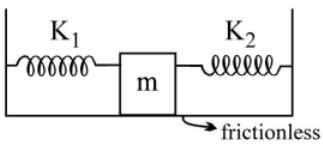
(A) average total energy per cycle of the motion is its maximum kinetic energy.

(B) average total energy per cycle of the motion is $\frac{1}{\sqrt{2}}$ times its maximum kinetic energy.

(C) root mean square velocity is $\frac{1}{\sqrt{2}}$ times its maximum velocity

(D) mean velocity is 1/2 of maximum velocity.

Q7. Two springs with negligible masses and force constant of $K_1 = 200\ Nm^{-1}$ and $K_2 = 160\ Nm^{-1}$ are attached to the block of mass m = 10 kg as shown in the figure. Initially the block is at rest, at the equilibrium position in which both springs are neither stretched nor compressed. At time t = 0, a sharp impulse of 50 Ns is given to the block with a hammer.



(A) Period of oscillations for the mass m is $\frac{\pi}{3}$ s.

(B) Maximum velocity of the mass m during its oscillation is 5 ms^{-1}.

(C) Data are insufficient to determine maximum velocity.

(D) Amplitude of oscillation is 0.42 m.

Q8. A mass of 0.2kg is attached to the lower end of a massless spring of force-constant 200 N/m, the upper end of which is fixed to a rigid support. Which of the following statements is/are true?

(A) In equilibrium, the spring will be stretched by 1cm.

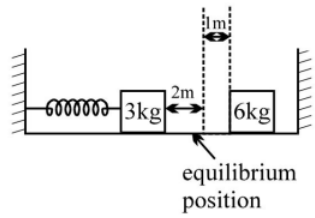
(B) If the mass is raised till the spring is unstretched state and then released, it will go down by 2cm before moving upwards.

(C) The frequency of oscillation will be nearly 5 Hz.

(D) If the system is taken to the moon, the frequency of oscillation will be the same as on the earth.

Q9. Two blocks of masses 3 kg and 6 kg rest on a horizontal smooth surface.

The 3 kg block is attached to a spring with a force constant k = 900 Nm^{-1} which is compressed 2 m from beyond the equilibrium position. The 6 kg mass is at rest at 1m from mean position. 3kg mass strikes the 6 kg mass and the two stick together.



(A) velocity of the combined masses immediately after the collision is 10 ms^{-1}

(B) velocity of the combined masses immediately after the collision is 5 ms^{-1}

(C) Amplitude of the resulting oscillation is $\sqrt{2}$ m

(D) Amplitude of the resulting oscillation is $\sqrt{5}/2$ m.

Q10. A particle moves in the x-y plane according to the equation, $\vec{r} = (\hat{i} + 2\hat{j})A\cos\omega t$. The motion of the particle is-

(A) on a straight line

(B) on an ellipse

(C) periodic

(D) simple harmonic

Q11. A particle is subjected to two simple harmonic motions along x and y directions according to, $x = 3\sin 100\pi t$; $y = 4\sin 100\pi t$.

(A) Motion of particle will be on ellipse traversing it in clockwise direction

(B) Motion of particle will be on a straight line with slope 4/3

(C) Motion will be simple harmonic motion with amplitude 5

(D) Phase difference between two motions is $\pi/2$

Q12. A particle free to move along the x-axis has potential energy given by :

$U(x) = k[1 - \exp(-x^2)]$ for $-\infty \le x \le +\infty$, where k is a positive constant of appropriate dimensions.

Then :-

(A) at points away from the origin, the particle is in unstable equilibrium

(B) for any finite non-zero value of x, there is a force directed away from the origin

(C) if its total mechanical energy is $\frac{k}{2}$, it has its minimum kinetic energy at the origin

(D) for small displacements from x = 0, the motion is simple harmonic.

Q13. A simple pendulum has time period T_1. The point of suspension is now moved upward according to the relation $y = Kt^2$, $(K = 1\ m/s^2)$ where y is the vertical displacement. The time period now becomes T_2. The ratio of $\frac{T_1^2}{T_2^2}$ is : $(g = 10\ m/s^2)$

(A) $\frac{6}{5}$ (B) $\frac{5}{6}$

(C) 1 (D) $\frac{4}{5}$

PARAGRAPH TYPE

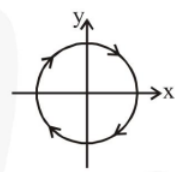
*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for question no. 14 and 15

Lissajous figures are produced by superposition of 2 SHM's in mutually perpendicular directions. If both SHM have same frequency, the lissajous figure is simple. It may be a circle, ellipse or a straight line depending on the phase difference between two SHMs. Interesting case occurs when one of the SHM is at double frequency of another. Suppose a body executes SHM vertically with frequency f_0, but horizontally with a frequency $2f_0$ and is initially at A, mean position of vertical as well as horizontal SHM. It can be seen to trace out a figure of 8 in space as shown. Since horizontal SHM has half the time period, it executes two horizontal oscillations by the time it completes a vertical oscillation.



Q14. A lissajous figure as shown here can be produced by



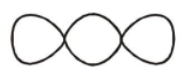
(A) $x = A \sin \omega t\ ;\ y = A \cos \omega t$

(B) $x = A \cos \omega t\ ;\ y = A \sin \omega t$

(C) $x = A \sin \omega t\ ;\ y = A \sin\left(\omega t + \frac{\pi}{4}\right)$

(D) $x = A \sin \omega t\ ;\ y = A \sin\left(\omega t + \frac{3\pi}{4}\right)$

Q15. For the Lissajous figure shown here, the frequency in vertical direction is _____ times the frequency in horizontal direction.

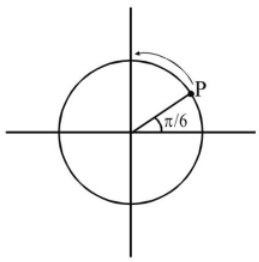
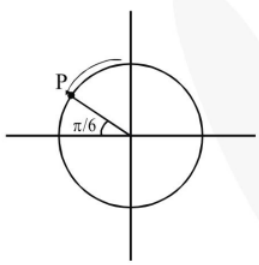
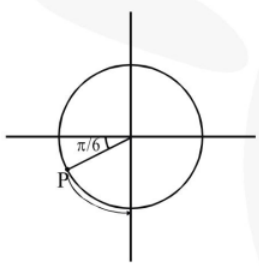
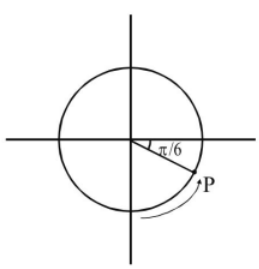


(A) 3 (B) $\frac{1}{3}$ (C) 2 (D) $\frac{1}{2}$

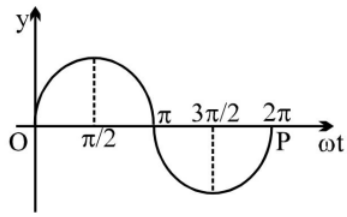
MATCH THE COLUMN TYPE

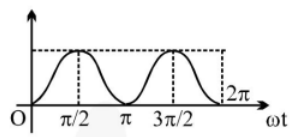
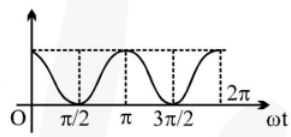
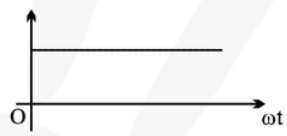
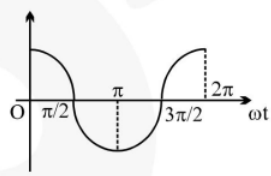
*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q16. All the phasor diagrams in column I are representing SHM, which has same time period, same amplitude and same angular frequency. The phasor moves in anticlockwise direction and projections on vertical diameter of reference circle represent SHM.

Column-I	**Column-II**
(A) 	(P) Equilibrium position is reached earliest.
(B) 	(Q) Acceleration is towards mean position.
(C) 	(R) Speed is increasing.
(D) 	(S) Velocity will reversed its direction in T/4 time
	(T) It crosses mean position in time interval T/6.

Q17. The graph plotted between phase angle (ϕ) and displacement of a particle from equilibrium position (y) is a sinusoidal curve as shown below. Then the best matching is :-



Column-I	Column-II
(A) K.E. versus phase angle curve	(P) 
(B) P.E. versus phase angle curve	(Q) 
(C) T.E. versus phase angle curve	(R) 
(D) Velocity versus phase angle curve	(S) 

SUBJECTIVE TYPE

Q18. Periodic time of oscillation T_1 is obtained when a mass is suspended from a spring if another spring is used with same mass then periodic time of oscillation is T_2. Now if this mass is suspended from series combination of above springs then calculate the time period.

Q19. A block is on a horizontal slab which is moving horizontally and executing S.H.M. The coefficient of static friction between block and slab is μ. If block is not separated from slab then determine angular frequency of oscillation.

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	B	A	C	A	B	D	A	A	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**		
C	A	C	A	B	A	D	B		

Ex-1A

1. C,D **2.** A **3.** B, C **4.** A,C **5.** B,C,D

6. A, C **7.** 3.2 kg **8.** 1 : 1, $-\frac{\pi}{12}$

10. (A) 0.25 Hz (B) $\frac{5}{\sqrt{2}}$ cm (C) 12.3 cm/s^2 (D) ±6.28 cm/s **11.** $x = \frac{\sqrt{3}A}{2}$

12. (a) $\frac{3}{4}, \frac{1}{4}$ (b) $\pm\frac{A}{\sqrt{2}}$ **13.** $k_1 = k\left[1+\frac{n_2}{n_1}\right], k_2 = k\left[1+\frac{n_1}{n_2}\right]$

14. $\frac{2mg}{k}$ **15.** $\ell_m = \frac{\ell_e}{6} = \frac{1}{6}$ m

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	A	B	C	A	A	D	A	C	C
11	**12**	**13**	**14**	**15**					
A	B	B	B	A					

Ex-2A

1. A,B,C,D **2.** A, C, D **3.** A,B,C,D **4.** A,B,C **5.** A,B,C

6. A,C **7.** A,B **8.** A,B,C,D **9.** A,C **10.** C,D

11. B,C **12.** D **13.** A **14.** A **15.** A

16. (A) -Q, S ; (B) -P,Q,R,T ; (C) -Q,S ; (D) -P,Q,R,T

17. (A) -Q, (B) -P, (C) -R, (D) -S **18.** $\sqrt{T_1^2 + T_2^2}$ **19.** $\omega \le \sqrt{\frac{\mu g}{A}}$

-: Important Notes & Formulas :-